LESSON PLAN TEMPLATE

PREPARE

Objectives: By the end of the lesson the students will be able to… (how will you know they know?)

1. Students will be able to apply Green's Theorem to compute the line integral along the boundary of a region in R^2.
2. Students will be able to apply Green's Theorem to compute the area of a region in R^2 by computing the corresponding line integral along the boundary.
3. Students will be able to apply Green's Theorem to compute the flux across the boundary of a region.

Materials:

- Tablet, lecture notes (attached)

Personal goals (what I'm working on in my teaching):

1. Time management
2. Effective use of tablet with *prewritten notes*
3. Clarity of explaining steps with prewritten notes
4. Eye contacts with students; inviting vibe of classroom

INSTRUCT

Intro:

Brief self-introduction:

- Name, current area of interest (connection to multivariable calculus/Green's Theorem!)
- Thank for the opportunity of practice teaching

Bridge-in to create interest and motivation to learn:

- Clark introduced the formula of Green's Theorem in Wednesday's class. I will do a quick recap and apply the theorem to examples. We will see the power of Green's Theorem in real-world problems that allow us to compute the area of and the flux across very irregular regions in R^2.

State learning objectives and connections:

- Practice Green's Theorem in problems.

Learning Activities: What you will do, what they will do, check for understanding (assess). Connect activities.

Activity: Review of Green's Theorem (~5min)

Activity: Warm-up exercises of using Green's Theorem to compute line integrals (~10min)
Assess: Invite questions
Connect: Compare two ways of computing Line integral

Activity: Apply Green's Theorem to compute the area of a region in R^2 (~8min)
Assess: Invite questions
Connect: Compare two ways of computing the area of a region in R^2

Activity: Apply Green's Theorem to compute the flux across the boundary of a region in R^2 (~8min)
Assess: Invite questions
Connect: Flux across a surface and divergence of a function

Closing:

Summarize lesson (big ideas to take away from the examples):

- To compute a line integral along a curve, think about the region "bounded" by the curve (if any)
- Compute the area of a region in R^2 via the corresponding line integral along the boundary
- 2-D divergence theorem

LESSON PLANNING CHECKLIST

Learning Objective(s) – your objectives…

- ☐ clearly state what you expect students will be able to do by the end of the instructional unit.
- ☐ reflect the level of learning that you want to see.
- ☐ express what the learners will do (rather than what you, as the teacher, will do).
- ☐ explain to learners what "good" looks like.

Review and Bridge-In – your review and bridge-in…

- ☐ create relevance and interest in the topic/subject of instruction.
- ☐ generate interest and focus in the learners.
- ☐ make links between previous instruction and the current focus of learning.
- ☐ draw on what learners are aware of and what they already know about the topic.

Teaching Points/Concepts - your concepts…

- ☐ clearly convey the teaching points to learner.
- ☐ cover what students need to know to successfully achieve the objectives.

Learning Activities (Application) – your learning activities…

- ☐ provide learners with supportive, step by step opportunities to learn and practice using the content to achieve the objectives.
- ☐ reinforce the learning necessary to achieve the objectives.
- ☐ allow students to learn in a safe environment.

Review and Formative Assessment – your assessment…

- ☐ provides feedback to both the teacher and the students on the learning that has occurred.
- ☐ measures achievement of the objectives.
- ☐ focuses students on the learning that they will be responsible for demonstrating.
- ☐ reinforces the key elements of learning.

Closing and Link to Summative Evaluation – the closing…

- ☐ shows students explicitly how evaluation aligns with the objectives and teaching/learning activities.
- ☐ link learning to course outcomes and summative evaluations.

At the beginning, I plan to:

1. Use a bridge-in to capture students' interest and motivate them to learn ..... ❑
2. Make the learning outcomes clear. ..... ❑
3. Assess prior learning and student expectations ..... ❑

In the middle, I plan to:

1. Use strategies to actively involve students in the learning process ..... ❑
2. Use a variety of media to illustrate concepts and processes ..... ❑
3. Ensure that the lesson flows easily and logically ..... ❑
4. Ensure that students are learning material that is meaningful and new ..... ❑
5. Provide opportunities for practice and feedback ..... ❑
6. Review and build on related material ..... ❑

At the end, I plan to:

Provide the proper closure students find important. To do this, I will:

1. Assess what students have learned ..... ❑
2. Summarize the lesson ..... ❑
3. Connect the lesson to real life and/or the next lesson ..... ❑

Examples of applying Green's Thm

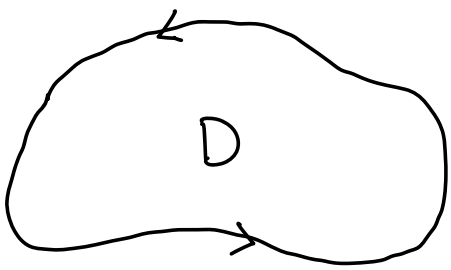
Green's Thm.

For a domain $D \subset \mathbb{R}^2$ and $\vec{F} = (P, Q)$:

$$\int_{\partial D} \vec{F} \cdot d\vec{x} = \iint_D (\operatorname{grn} \vec{F})\, dA$$

oriented.

$$= \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$



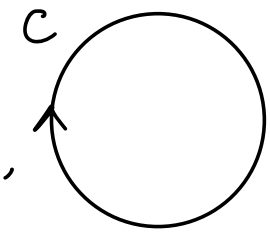
Circulating $\vec{F}$ around the boundary of D

= Accumulating the "circulation density" of $\vec{F}$ inside D.

Green's Thm

$$\int_{\partial D} \vec{F}\cdot d\vec{x} = \iint_D (\operatorname{grn}\vec{F})\,dA = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

Ex 1



C is the clockwise unit circle,

$\vec{F} = (e^x + y,\ \tan y + 3x)$. Compute:

$$\int_C \vec{F}\cdot d\vec{x}.$$

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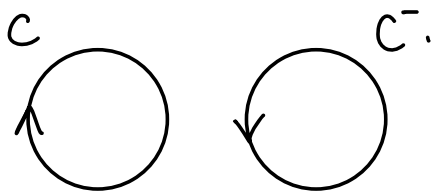
① Old way:

Parametrize C and F, *&^%^! ...

② Now !:



Consider the unit disc bounded by C. Its oriented boundary is C^-!

$$\int_C \vec{F}\cdot d\vec{x} \underset{\text{property of line integral}}{=} -\int_{C^-} \vec{F}\cdot d\vec{x} = -\int_{\partial D} \vec{F}\cdot d\vec{x} \underset{\text{Green's Thm}}{=} -\iint_D \operatorname{grn}\vec{F}\,dA.$$

$$= -\iint_D \left(\frac{\partial(\tan y + 3x)}{\partial x} - \frac{\partial(e^x + y)}{\partial y} \right) dA$$

$$= -\iint_D (3-1)\, dA$$

$$= -2 \cdot \text{Area}(D)$$

$$= -2\pi$$

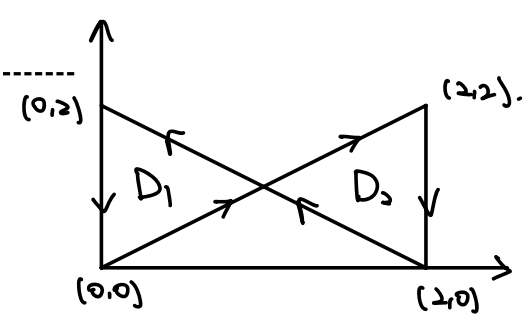
Idea:

Given a line integral along a curve C, think about the region "bounded" by C (if any!).

Ex 2.

C is made up of line segments from $(0,0)$ to $(2,2)$ to $(2,0)$ to $(0,2)$ to $(0,0)$, and $F = (3x-4y, 5x-7y)$.

Compute $\int_C \vec{F}\, d\vec{x}$.



Note: $C =$ $+$

$= \partial D_1 + (\partial D_2)^-$

So: $\int_C \vec{F} \cdot d\vec{x} = \int_{\partial D_1} \vec{F} \cdot d\vec{x} + \int_{\partial D_2} \vec{F} \cdot d\vec{x}$

$= \int_{\partial D_1} \vec{F} \cdot d\vec{x} + \int_{(\partial D_2)^-} \vec{F} \cdot d\vec{x}.$

$$= \int_{\partial D_1} \vec{F} \cdot d\vec{x} - \int_{\partial D_2} \vec{F} \cdot d\vec{x}.$$

$$= \iint_{D_1} \operatorname{grn} \vec{F} \cdot dA - \iint_{D_2} \operatorname{grn} \vec{F} \cdot dA$$

$$= \iint_{D_1} \left(\frac{\partial(5x-7y)}{\partial x} - \frac{\partial(3x-4y)}{\partial y} \right) dA$$

$$- \iint_{D_2} \left(\frac{\partial(5x-7y)}{\partial x} - \frac{\partial(3x-4y)}{\partial y} \right) dA.$$

$$= \iint_{D_1} 9 \, dA - \iint_{D_2} 9 \, dA$$

$$= 9 \cdot \text{Area}(D_1) - 9 \cdot \text{Area}(\triangle)$$

$$= 0$$

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2. Applications of Green's Thm
(in a more signifcant way...)

$\boxed{\text{Area.}}$

As seen above, line integral along a curve is directly related to the area of the region bounded by the curve if $\operatorname{grn} \vec{F}$ is constant...

! This also works BACKWARDS:

$\boxed{Q}$: Can we compute the area of a region in $\mathbb{R}^2$ bounded by a curve by integrating <u>some function F</u>? along the curve?

$\boxed{A}$ Consider $F = (0, x)$!

$$\text{Area}(D) = \iint_D 1\, dA = \iint_D \left(\frac{\partial x}{\partial x} - \frac{\partial 0}{\partial y}\right) dA.$$

$$= \iint_D \operatorname{grad}\begin{pmatrix}0\\x\end{pmatrix} dA$$

Green's Thm $= \int_{\partial D} \begin{pmatrix}0\\x\end{pmatrix} d\vec{x}$

Defn of line integral $= \int_{\partial D} x\, dy$

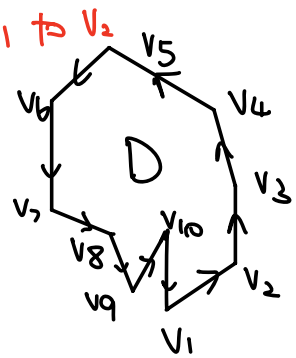
Now, if D is a polygon with oriented vertices $v_1 \cdots v_n$

∂D consists of straight line segment L_1 from v_1 to v_2

L_2 from v_2 to v_3

$\vdots$

L_n from v_n to v_1



Then:

$$\text{Area}(D) = \int_{\partial D} x\, dy = \sum_{i=1}^{n} \int_{L_i} x\, dy$$

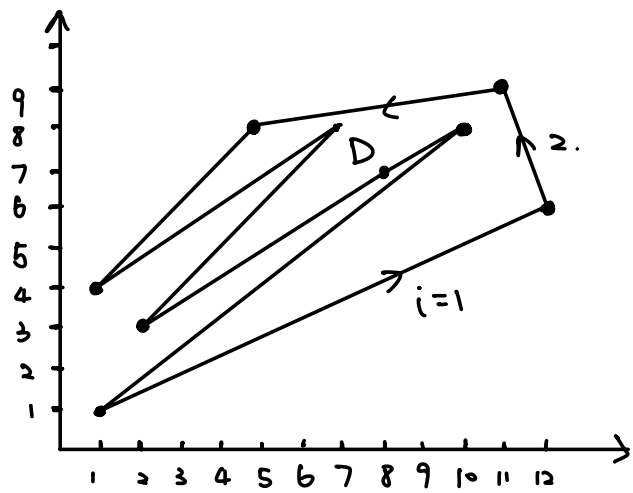
You can check for any straight line segment L from (x_1, y_1) to (x_2, y_2), we have

$$\int_L x\, dy = \left(\frac{x_1 + x_2}{2}\right)(y_2 - y_1) = \bar{x}\, \Delta y$$

<u>Conclusion</u>: $\text{Area}(D) = \sum_{i=1}^{n} \left(\frac{x_i + x_{i+1}}{2}\right)(y_{i+1} - y_i)$

if D is a polygon $(x_{n+1} = x_1,\ y_{n+1} = y_1)$.

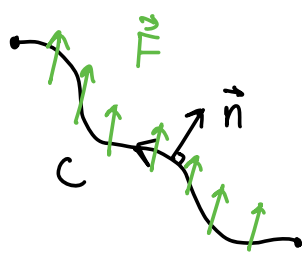
Ex.



i	$\bar{x}$	Δy	$\bar{x}\Delta y$
1	$13/2$	5	$65/2$
2	$23/2$	3	$69/2$
3	8	-1	-8
4	3	-4	-12
5	4	4	16
6	$9/2$	-5	$-45/2$
7	5	4	20
8	9	1	9
9	$11/2$	7	$-77/2$

Σ = area $= 19$.

Flux through a curve



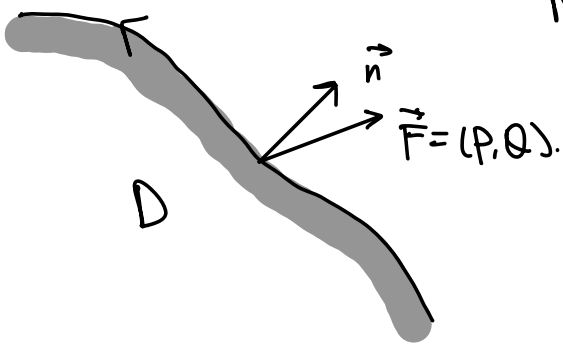
Let C be a simple, oriented curve in $\mathbb{R}^2$. $\vec{F}$ a function. $\vec{n}$ the unit normal to C obtained by rotating the unit tangent vector clockwise by $90°$. The flux across C is line integral

$$\int_C \vec{F}\cdot\vec{n}\, ds.$$

intuition:

$\vec{F}$ = velocity vector field of a fluid

Φ = rate of fluid transported across C per unit time.



Now, if the oriented curve C is the boundary ∂D of some region D, you can check that

$\vec{n}$ is pointing outwards

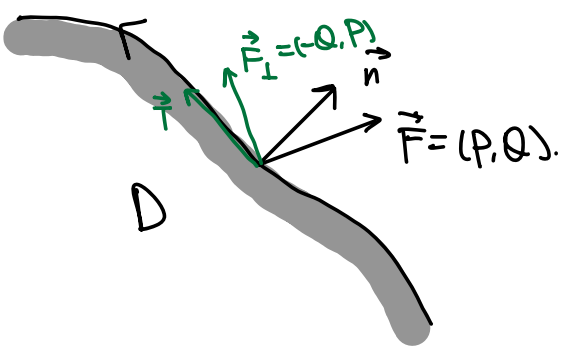
and so the flux across C is

the rate of fluid escaping from D

$\boxed{Q}$: How do use the (extra) condition that ∂D is the boundary of a region D to compute the flux across ∂D?

$\boxed{A}$: Green's Thm !

Rotate $\vec{F}, \vec{n}$ counterclockwise by $90°$ to obtain $\vec{F}_\perp, \vec{T}$



Know from linear algebra that

$\vec{F}_\perp = (-Q, P)$ & $\vec{F} \cdot \vec{n} = \vec{F}_\perp \cdot \vec{T}$

Then:

$$\Phi = \int_{\partial D} \vec{F} \cdot \vec{n} \, ds$$

$$= \int_{\partial D} \vec{F}_\perp \cdot \vec{T} \, ds$$

$$= \int_{\partial D} \vec{F}_\perp \cdot dx$$

$$\overset{\text{Green's thm}}{=} \iint_D \operatorname{div}(\vec{F}_\perp) \, dA$$

$$= \iint_D \left(\frac{\partial P}{\partial x} - \frac{\partial(-Q)}{\partial y} \right) dA$$

$$= \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA$$

$$= \iint_D \nabla \cdot \vec{F} \, dA.$$

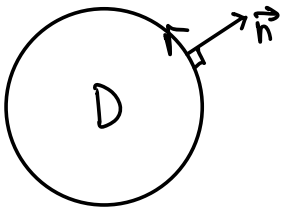
Conclusion (2D divergence Thm):

$$\int_{\partial D} \vec{F} \cdot \vec{n} \, ds = \iint_D \nabla \cdot \vec{F} \, dA.$$

Ex: Compute the flux $\int_C \vec{F} \cdot \vec{n} \, ds$ of $\vec{F} = (2x + e^y, \csc x - 5y)$ outward through the unit circle C.

In other words: want to compute the flux across the circle, oriented counterclockwise.

C, with this orientation, is ∂D, where D is the unit disk. So:



$$\int_C \vec{F} \cdot \vec{n} \, ds = \int_{\partial D} \vec{F} \cdot \vec{n} \, ds \overset{\text{2D divergence}}{=} \iint_D \nabla \cdot \vec{F} \, dA$$

$$= \iint_D \left(\frac{\partial (2x + e^y)}{\partial x} + \frac{\partial (\csc x - 5y)}{\partial y} \right) dA$$

$$= \iint_D (2 - 5) \, dA$$

$$= -3 \cdot \text{Area}(D)$$

$$= -3\pi.$$